

ANSWER BOOK

I AGREE TO ABIDE BY THE SENATE POLICY ON CONDUCT OF ASSIGNMENTS, MID-TERM EXAM, AND FINAL EXAM

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Assignment on **Fuzzy Logic and Reasoning**

Q2b. Solution:

Rule 1 - Consequent Membership Function:

$$\begin{aligned} B1 &= \text{TrapMF}(y, [0, 2, 4, 6]) \\ \text{Centroid} &: (6 - 0) / 2 = 3 \\ \text{Area} &: [(2 + 6) / 2] * 1 = 4 \end{aligned}$$

Rule 2 - Consequent Membership Function:

$$\begin{aligned} B2 &= \text{TriMF}(y, [4, 6, 8]) \\ \text{Centroid} &: (4 + 8) / 2 = 6 \\ \text{Area} &: (8 - 4) / 2 = 2 \end{aligned}$$

Rule 3 - Consequent Membership Function:

$$\begin{aligned} B3 &= \text{TrapMF}(y, [6, 8, 10, 12]) \\ \text{Centroid} &: (6 + 12) / 2 = 9 \\ \text{Area} &: [((12 - 6) + (10 - 8)) / 2] * 1 = 4 \end{aligned}$$

$$Y = \frac{(0.2 * 4 * 3) + (0.5 * 2 * 6) + (0.3 * 4 * 9)}{(0.2 * 4) + (0.5 * 2) + (0.3 * 4)}$$

$$Y = \frac{19.2}{3} = \mathbf{6.4000}$$